THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM THREE MARCH MID–TERM EXAMINATIONS

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 3:00 Hours **Year: 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Name in Full** on every page of your answer booklet(s).

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KSS MARCH MID–TERM 2024

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) Which of the following is **NOT** a contribution of biology in the field of forestry?

A Increased desertification
B Fast maturing plants
C Varieties of trees that grow well in arid areas
D Solutions to plant diseases and pests
E Determination of soil health and ways to improve it

(ii) Information from the sense organ in form of electrical signals is transmitted towards the central nervous system (CNS) by:

A Afferent neuron
B Intermediate neuron
C Motor neuron
D Nerve cell
E Relay neuron

(iii) A red blood cell burst if it is left in hypotonic solution for too long since the solution has:

A Low water potential as compared to the intracellular environment
B High water potential as compared to the intracellular environment
C High solute potential as compared to the intracellular environment
D The same water potential as the intracellular environment
E The same solute potential as the intracellular environment

(iv) One of the features of dicot plant leaves is that:

A They have parallel venation
B They are short and broad
C Their seeds have two cotyledons
D Their vascular bundles are scattered
E They have a tap root system

(v) A vector for malaria is:

A Plasmodium
B Stagnant water
C Mosquito
D Insecticides
E Bacteria

(vi) Examples of specialized cells in plants and animals respectively include:

A Chloroplast and red blood cell
B Stomata and white blood cell
C Palisade and sperm cell
D Root hair and eukaryotic cell
E Lenticel and platelets

(vii) Some vitamins should be taken regularly since they cannot be stored in the body because:

A They are not absorbed by the body, for example vitamin K
B They are soluble in fat, for example vitamin E
C They are not easily decomposed by heat, for example vitamin A
D They are soluble in water, for example vitamin B
E They are easily oxidized during respiration, for example vitamin D

(viii) When roots of a plant grow away from the source of light they are said to undergo:

A Positive phototropism
B Positive geotropism
C Negative geotropism
D Negative phototropism
E Facilitated hydrotropism

(ix) During a practical session, Lightness and Yulia were given lime water *(calcium hydroxide solution)* and asked to breath out into it. After many attempts, the solution changed into a milky colour. This is because:

A Exhaled air contains oxygen gas which is milky in colour
B Exhaled air contains carbon dioxide which is milky in colour
C Exhaled air is warm and facilitate decomposition of lime water
D Exhaled air contains moisture which forms a milky colour with lime water
E Exhaled air contains carbon dioxide which forms calcium carbonate with lime water

(x) Which of the following symbols is most likely to be found on a bottle of Uhai drinking water?

A 

B 

C 

D 

E 

2. Match the item in List A with the response in List B by writing the letter of the correct response besides the item number in the answer booklet(s) provided.

LIST A	LIST B
(i) Smooth muscle	A A muscle that is able to contract without suffering fatigue and they are not under the influence of nervous system
(ii) Cardiac muscle	B A muscle with ability to pull two limb bones closer
(iii) Flexor	C Slender threads running along the length of the muscle fibre
(iv) Skeletal muscle	D A cell membrane present in muscle cells
(v) Sarcolemma	E A muscle made up of long and tapered end cells found in most of internal organs such as uterus, blood vessels and sperm duct
(vi) Myofibrils	F A muscle made up of long fibres that cover the bones
	G A muscle with ability to pull two limb bones apart

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Give three (3) differences between cell wall and cell membrane basing on the criteria provided:

S/N	CRITERIA	CELL WALL	CELL MEMBRANE
(i)	Permeability		
(ii)	Composition		
(iii)	State of living		

(b) With the aid of a labeled diagram, explain the state of your upper arm when the arm is bent.

4. (a) State the function of each of the following:
   (i) Beam balance
   (ii) Petri dish
   (iii) Hinge screw on the microscope
   (iv) Liniment in the first aid kit
   (b) Briefly explain the role played by the following parts of the digestive system:
   (i) Rectum
   (ii) Duodenum
   (c) How is the ileum adapted to ensure maximum absorption? Give three (3) points.

5. (a) What is ecology?
   (b) Kindoroko secondary school garden comprises of beetles, grasses, spring water, vulture, fertile soil, lizards, fresh air, locusts and snakes.
   (i) Identify all abiotic components present in Kindoroko secondary school garden.
   (ii) Construct a food web accommodating all organisms found in the garden.
   (iii) Name the important group of organisms missing in this ecosystem and state its role.

6. (a) Study the diagram below and answer the questions that follow.

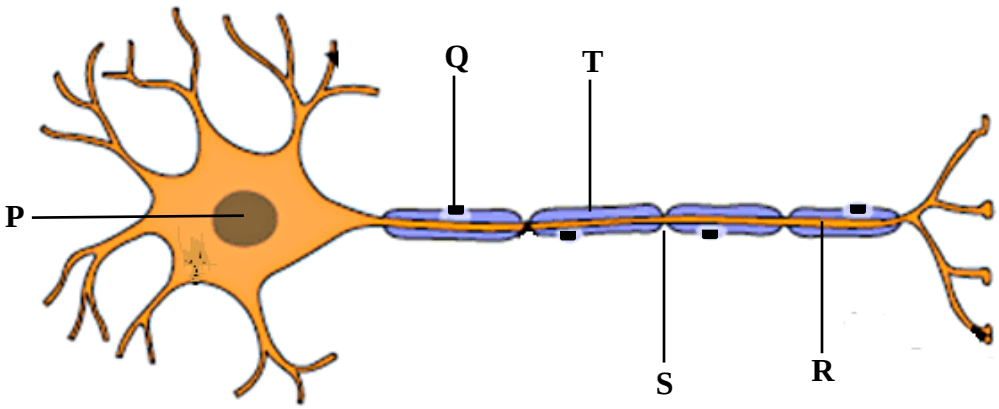


Figure 1

   (i) Give the name of the structure presented in figure 1 above.
   (ii) Name the parts labeled P, Q and R.
   (iii) State the function of the parts S and T.
   (b) Explain three (3) ways in which the brain is adapted to its functions.

7. (a) Moss plants do not grow on a dry land; they are usually found in wet and moist places. Account for this observation.

   (b) What are possible signs that a person is suffering from internal bleeding? Give three (3) points.

   (c) Name the structure used by the following organisms for gaseous exchange:
   (i) Spider
   (ii) Amoeba
   (iii) Young tadpole
   (iv) Rat

8. (a) Jackline and Glory were interested to study fungi in details. They collected many organisms from all phyla of this kingdom but their study was mainly based on phylum basidiomycota. They were surprised to see that none of the organisms they collected was green in colour.
   (i) What name should be given to Jackline and Glory basing on their field of study?
   (ii) In comparison to plants, why fungi are not green in colour?
   (iii) Why are fungi not placed in the first trophic level in any feeding relationship?
   (iii) Draw a labeled diagram of any representative organism from the phylum that they studied.

   (b) How can girls avoid unpleasant smell and diseases during adolescence? Explain three (3) points.

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. Our bodies possess a special organ for pumping blood. Explain six (6) features that aid this organ to perform its function efficiently.

10. Plants are able to utilize solar energy, carbon dioxide and water to make their own carbohydrates. This process is said to be very important not only to plants but to the environment at large.

    (a) Write a chemical equation for the process described above.

    (b) Write an essay on five (5) significance of this process in nature.

11. Why do you think scientists were convinced to believe that bacteria are living things despite of being too small to be seen with naked eyes? Explain by giving six (6) points.

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